

# Joydeep Biswas

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## Current Appointment

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Associate Professor, Computer Science Department, The University of Texas at Austin  
Associate Director, Texas Robotics, The University of Texas at Austin

Visiting Professor, NVIDIA

## Education

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| 2014 | Ph.D. in Robotics              | Carnegie Mellon University             |
| 2010 | M.S. in Robotics               | Carnegie Mellon University             |
| 2008 | B.Tech. in Engineering Physics | Indian Institute of Technology, Bombay |

## Achievements and Awards

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| 2026 | Best Paper Award Finalist, ICRA 2026, <i>GuideTWSI</i>                                                             |
| 2026 | Best Paper Honourable Mention, HRI 2026, <i>GuideNav</i>                                                           |
| 2025 | Third Place, RoboCup 2025 Humanoid League (Adult Size), <i>UT Austin Villa</i>                                     |
| 2025 | Best Paper Runner-up, RSS 2025 Workshop on Resilient Off-road Autonomous Robotics, <i>OverSeeC</i>                 |
| 2024 | Best Paper Award, ACM CHI Conference on Human Factors in Computing Systems, <i>Informed Guide Dog Robot Design</i> |
| 2024 | JP Morgan Faculty Research Award                                                                                   |
| 2022 | 1st Place, Benchmark Autonomous Robot Navigation (BARN) Challenge, ICRA 2022                                       |
| 2021 | NSF CAREER Award                                                                                                   |
| 2019 | Student Best Poster Award, Northrop Grumman University Symposium                                                   |
| 2019 | IJCAI Early Career Spotlight                                                                                       |
| 2019 | Amazon Research Award                                                                                              |
| 2018 | JP Morgan AI Faculty Research Award                                                                                |
| 2018 | Best Demo Award, AAMAS 2018                                                                                        |
| 2018 | 5th place, RoboCup 2018 Small Size League, <i>UMass MinuteBots</i> , Faculty Team Leader                           |
| 2017 | Lower Bracket 1st place, RoboCup 2017 Small Size League, <i>UMass MinuteBots</i> , Faculty Team Leader             |
| 2015 | Siebel Scholar, Class of 2015                                                                                      |
| 2015 | 1st place, RoboCup 2015 Small Size League, <i>CMDragons</i> , Student Team Leader                                  |
| 2014 | 2nd place, RoboCup 2014 Small Size League, <i>CMDragons</i> , Student Team Leader                                  |
| 2013 | 2nd place, RoboCup 2013 Small Size League, <i>CMDragons</i> , Student Team Leader                                  |
| 2010 | 2nd place, RoboCup 2010 Small Size League, <i>CMDragons</i> , Team Member                                          |

## Employment History

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|----------------|-----------------------------|--------------------------------------------------------------------------------------------|
| 2023 - Present | Associate Professor         | Computer Science Department, University of Texas at Austin, TX, USA                        |
| 2017 - 2023    | Technical Advisor           | Consumer Robotics, Amazon Lab 126                                                          |
| 2019 - 2023    | Assistant Professor         | Computer Science Department, University of Texas at Austin, TX, USA                        |
| 2019 - 2025    | Adjunct Assistant Professor | College of Information and Computer Sciences, University of Massachusetts Amherst, MA, USA |
| 2015 - 2019    | Assistant Professor         | College of Information and Computer Sciences, University of Massachusetts Amherst, MA, USA |
| 2015           | Post-Doctoral Fellow        | Computer Science Department, Carnegie Mellon University, Pittsburgh, PA, USA               |
| 2012           | Summer Intern               | Google Research, Mountain View, CA, USA                                                    |
| 2010           | Summer Intern               | Intel Research, Pittsburgh, PA, USA                                                        |

## Funding

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### Federal Funding

#### **NIH Award “*Human-Centered Development of Guide-dog Robots*”**

Role: Co-PI.

PI: Donghyun Kim (UMass Amherst).

Co-PIs: Ivan Lee (UMass Amherst), Nicholas Giudice (University of Maine).

Period: October 2025 – September 2028.

#### **NSF Award “*Collaborative Research: SLES: No Bad Surprises: Aligning Agent and Human Norms via Specification Refinements*”**

Role: Co-PI.

PI: Sandhya Saisubramanian (Oregon State University).

Co-PIs: Houssam Abbas (Oregon State University), Shlomo Zilberstein (UMass Amherst).

Period: October 2024 – September 2028.

#### **Army Research Laboratories Award “*SARA Sprint 3: Long-Range Offroad Introspective Driving*”**

Role: PI.

Period: January 2024 – December 2025.

#### **NSF Award “*FMitF: Track I: Program Synthesis for Robot Learning from Demonstrations*”**

Role: Co-PI.

PI: Isil Dillig.

Period: October 2023 – September 2027.

#### **NSF Award “*GCR: Community-Embedded Robotics: Understanding Sociotechnical Interactions with Long-term Autonomous Deployments*”**

Role: Co-PI.

PI: Luis Sentis.

Co-PIs: Elliott Hauser, Justin Hart, Keri Stephens.  
Period: October 2022 – September 2027.

**Army Research Laboratories Award “*Human-Guided Learning of Neuro-Symbolic Mission Execution Policies*”**

Role: PI.  
Co-PI: Isil Dillig.  
Period: September 2021 – January 2023.

**NSF Award “*NRT-AI: Convergent, Responsible, and Ethical Artificial Intelligence Training Experience for Roboticists*”**

Role: Co-PI.  
PI: Junfeng Jiao.  
Co-PIs: Luis Sentis, Justin Hart.  
Period: September 2021 – August 2026.

**NSF Award “*CAREER: Robust Perception and Customization for Long-Term Autonomous Mobile Service Robots*”**

Role: PI.  
Period: April 2021 – March 2026.

**NSF Award “*RI: Medium: Introspective Perception and Planning for Long-Term Autonomy*”**

Role: PI.  
Co-PI: Shlomo Zilberstein (UMass)  
Period: July 2020 – June 2023.

**NSF Award “*SHF: Small: Interactive Synthesis and Repair For Robot Programs*”**

Role: Co-PI.  
PI: Arjun Guha (UMass)  
Period: June 2020 – May 2023.

**DARPA Award “*Advancing Learning via Probabilistic Causal Analysis for Competency Awareness*”**

Role: Co-PI.  
PI: Charles River Analytics Co-PI: David Jensen (UMass).  
Period: October 2019 – September 2022.

**Army Futures Command Robotics Center of Excellence “*Persistent Fully Autonomous Multi-Robot Tactics in Complex Environments*”**

Role: Co-PI.  
PI: Peter Stone Co-PIs: Luis Sentis, Justin Hart  
Period: October 2019 – December 2022.

**NSF Award “*S&AS: FND: Reliable Semi-Autonomy with Diminishing Reliance on Humans*”**

Role: Co-PI.  
PI: Shlomo Zilberstein (UMass)  
Period: September 2017 – August 2020.

**DARPA Award “*Intelligent Model-Based Adaptation for Mobile Robotics*”**

Role: Co-PI.  
PI: Jonathan Aldrich (CMU) Co-PIs: David Garlan (CMU), Manuela Veloso (CMU), Christian Kaestner (CMU), Claire Le Gouess (CMU).  
Period: November 2015 – November 2019.

## **Competitive Industry Awards**

**Amazon Faculty Research Award, 2024**

Role: PI.  
Collaborators: Roberto Martin-Martin  
Period: September 2025 – August 2026.

**JP Morgan Faculty Research Award, 2023**

Role: PI.  
Collaborators: Arjun Guha (Northeastern University)  
Period: September 2023 – August 2024.

**Northrop Grumman Mission Systems’ Research in Applications for Learning Machines (REALM) Consortium**

Role: Co-PI.  
PI: Shaoshuai Mou (Purdue) Co-PIs: Daniel A. DeLaurentis (Purdue), Bing Liu (UIC)  
Period: January 2019 – December 2021.

**JP Morgan AI Research Award, 2019**

Role: PI.  
Period: September 2019 – August 2020.

**Amazon Research Award, 2018**

Role: PI.  
Period: September 2019 – August 2020.

## **Teaching Experience**

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**Instructor, CS 304I, Fall 2025, Fall 2026: The Essentials of AI for Life and Society**

University-wide course, University of Texas at Austin

University-wide course, University of Texas at Austin

**Instructor, CS 393R, Fall 2020, Fall 2021, Spring 2024: Autonomous Robots**

Graduate course, University of Texas at Austin

**Instructor, CS 109, Fall 2023: The Essentials of AI for Life and Society**

University-wide course, University of Texas at Austin

**Instructor, CS 388U, Fall 2023: Planning, Search, and Reasoning Under Uncertainty**

Online MS course, University of Texas at Austin

**Instructor, CS 378H, Fall 2023: F1/10 Autonomous Driving – Honors**

Honors Undergraduate course, University of Texas at Austin

**Instructor, CS378/ME379M/ME397/ECE394J/ECE379K, Spring 2023: Connected Autonomous Electric Vehicles**

Undergraduate course, University of Texas at Austin

**Instructor, CS 393R, Spring 2023: Planning, Search, and Reasoning Under Uncertainty**

Graduate course, University of Texas at Austin

**Instructor, CS 378H, Spring 2022: F1/10 Autonomous Driving – Honors**

Honors Undergraduate course, University of Texas at Austin

**Instructor, CS 378F, Spring 2020, Spring 2021: F1/10 Autonomous Driving**

Undergraduate course, University of Texas at Austin

**Instructor, COMPSCI 220, Fall 2017, Fall 2018: Programming Methodology**

Undergraduate course, University of Massachusetts Amherst

**Instructor, COMPSCI 403, Fall 2016, Spring 2018 : Introduction To Robotics**

Undergraduate course, University of Massachusetts Amherst

**Instructor, COMPSCI 603, Spring 2016, Spring 2017, Spring 2019 : Robotics**

Graduate course, University of Massachusetts Amherst

**Instructor, COMPSCI 691BR, Spring 2017 : Building A Robot Soccer Team**

Graduate Seminar, University of Massachusetts Amherst

## **Invited Talks**

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Reading the Dirt: Learning What Matters (and What Doesn't) for Off-Road Navigation

*Keynote talk, RSS 2025 Workshop on Resilient Off-road Autonomous Robotics, June 2025*

Robots Without Boundaries: Embracing the Unseen and Tackling Novel Tasks

*Amazon Consumer Robotics Symposium, May 2025*

Robots Without Boundaries: Embracing the Unseen and Tackling Novel Tasks

*Keynote talk, Texas Regional Robotics Symposium, May 2025*

Teaching Robots To “Get It Right”

*Invited talk, Oregon State University, January 2025*

Teaching Robots To “Get It Right”

*Keynote talk, The 37th International FLAIRS Conference, May 2024*

From Social- To Everything- Navigation

*Workshop on Social Robot Navigation: Advances and Evaluation. IROS 2023, October 2023*

Towards Context-Aware Robot Navigation

*Workshop on Robotic Perception and Mapping: Frontier Vision & Learning Techniques: Advances and Evaluation. IROS 2023, October 2023*

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

*University of Maryland, October 2022*

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

*Northeastern University, October 2022*

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

*Wellesley College, October 2022*

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

*Samsung AI Center, NYC, June 2022*

Self-Supervised and User-Supervised Adaptation of Autonomous Robots

*JP Morgan AI Research Center, NYC, June 2022*

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

*Stanford University / Robotics Seminar, April 2022*

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

*University of Southern California / CS Colloquium, April 2022*

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

*Brown University / BigAI Talk, April 2022*

Motion Control and Visual Representation Learning for High-Speed Off-Road Driving

*University of Pennsylvania / FITenth Invited Lecture, April 2022*

Particle Filters for Mobile Robot Localization

*Wellesley College, February 2022*

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

*Nvidia, March 2021*

Anticipating and Avoiding Failures Using Introspective Perception and Physics-Informed Program Synthesis

*MIT Embodied Intelligence Seminar, February 2021*

Building Robots For Long-Term Autonomy, And Keeping Them Autonomous

*Yale University, April 2019*

The Quest for “Always-On” Autonomous Mobile Robots

*IJCAI 2019 Early Career Spotlight Talk, August 2019*

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous

*ICRA 2018 Workshop: Long-term Autonomy and Deployment of Intelligent Robots in the Real-world, May*

2018

Building Robots For Long-Term Autonomy, And Keeping Them Autonomous  
*Carnegie Mellon University*, March 2018

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous  
*Amazon*, November 2017

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous  
*IROS 2017 Workshop: Assistance and Service Robotics in a Human Environment*, September 2017

Autonomous Mobile Robot Perception for Changing Environments  
*ICRA 2016 Workshop: AI for Long-term Autonomy*, May 2016

Deploying Autonomous Service Mobile Robots, And Keeping Them Autonomous.  
*University of New Hampshire Robotics Seminar Series*, March 2016

The Quest for Robust, Reliable, Autonomous Mobile Robots.  
*Williams College Computer Science Department Colloquium*, November 2015

The Quest for Robust, Reliable, Autonomous Mobile Robots.  
*Vecna Robotics*, September 2015

The Quest for Robust, Reliable, Autonomous Mobile Robots.  
*University of Minnesota, Computer Science & Engineering*, April 2015

Vector Map-Based, Non-Markov Localization for Long-Term Deployment of Autonomous Mobile Robots  
*Google X*, April 2015

The Quest for Robust, Reliable, Autonomous Mobile Robots.  
*University of Massachusetts Amherst, School of Computer Science*, March 2015

The Quest for Robust, Reliable, Autonomous Mobile Robots.  
*University of Massachusetts Amherst, School of Computer Science*, March 2015

## Panels

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Panel Member, 2025 AAAI Presidential Panel on the Future of AI: <https://aaai.org/about-aaai/presidential-panel-on-the-future-of-ai-research/>

Panel Moderator, Ethics Aware Design of AI  
*2020 Global Analytics Summit: Ethics in AI, Texas McCombs*, November 2020

Discussion Panel, Record of Robotics at CMU Part II, A Live Interview with Manuela Veloso.  
*CMU Record of Robotics Series*, October 2020

Last Mile Autonomous Delivery Systems: A Live Webcast Demonstration, and Panel Discussion  
*UT Good Systems Webinar*, September 2020

The Call for an Accelerated Autonomy – Robotics on the Frontlines of a Crisis  
*Computing In Our New Normal: A UTCS Webinar*, May 2020

## Professional Service

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### Outreach Activities

- Science on Screen Series at Amherst Cinema, Amherst MA, 31 October 2018: Presented a introduction to “Christine” within the scientific context of actual self-driving cars. *The Radical Future of Self-Driving Cars*.
- SciTech Cafe, Northampton MA, 23 January 2017: Presented a scientific talk to a general public audience. “Where am I?” and Other Fundamental Questions Robots Think Long and Hard to Answer
- HolyokeCodes, Holyoke MA, 8–12 July 2019: Co-Organized with Arjun Guha, a week-long robotics workshop for high-school students with state-of-the-art soccer-playing robots that we used to compete with at RoboCup. We covered the basic robot sense-plan-act control cycle, computational geometry, and simple adversarial planning. Students implemented building blocks of increasingly complex robot behaviors, leading up to a robot soccer tournament.

### Significant Service Roles

- Local Chair, Conference on Robot Learning (CoRL): 2026
- Associate Program Chair, AAAI Conference on Artificial Intelligence (AAAI): 2026
- Senior Member Track Co-Chair, AAAI Conference on Artificial Intelligence (AAAI): 2025
- Associate Editor, The International Journal of Robotics Research: 2023 – present
- Robotics Area Chair, International Joint Conference on Artificial Intelligence (IJCAI): 2023
- Co-Organizer, Texas Regional Robotics Symposium: April 29, 2022
- Associate Editor, Elsevier Robotics and Autonomous Systems: 2019 – present
- RoboCup Federation Trustee: 2021 – Present
- Diversity and Inclusion Co-Chair, AAAI Conference on Artificial Intelligence (AAAI): 2022, 2023
- RoboCup Executive Committee, Small Size League: 2015 – 2021
- Robot Exhibitions Co-Chair, International Joint Conference on Artificial Intelligence (IJCAI): 2021
- RoboCup Symposium Co-Chair: 2020-2021
- Robotics Track Co-Chair, International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2019
- Associate Editor, IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): 2016



### **Senior Program Committee**

- AAAI Conference on Artificial Intelligence (AAAI): 2024
- AAAI Conference on Artificial Intelligence (AAAI): 2023
- International Joint Conference on Artificial Intelligence (IJCAI): 2022
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2022
- AAAI Conference on Artificial Intelligence (AAAI): 2020
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2021
- Autonomous Robots and Multirobot Systems Workshop (ARMS): 2016

### **Program Committee / Reviewer**

- Autonomous Robots and Multirobot Systems Workshop (ARMS): 2020
- AAAI Symposium on Educational Advances in Artificial Intelligence: 2021
- RoboCup Symposium: 2015, 2016, 2017, 2018, 2019
- AAAI Undergraduate Consortium: 2021
- IEEE/SICE International Symposium on System Integration (SII): 2019
- Robotics: Science and Systems (RSS): 2015, 2016, 2019
- International Symposium on Multi-Robot and Multi-Agent Systems (MRS): 2019
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2017, 2018
- International Conference on Automated Planning and Scheduling (ICAPS) : 2016, 2018, 2021
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS): 2009, 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020
- IEEE International Conference on Robotics and Automation (ICRA): 2010, 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 2020, 2021
- IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN): 2010
- IEEE Conference on Human-Robot Interaction (HRI): 2016
- International Joint Conference on Artificial Intelligence (IJCAI): 2016
- International Conference on Autonomous Agents and Multiagent Systems (AAMAS): 2016

### **Journal Reviewing**

- IEEE Robotics and Automation Letters (RA-L): 2017, 2018, 2019, 2020, 2021
- IEEE Robotics and Automation Magazine (IEEE-RAM): 2013, 2014, 2015, 2016, 2019
- IEEE Transactions on Robotics (T-RO): 2015, 2018, 2019, 2020
- International Journal of Robotics Research (IJRR): 2016, 2017
- International Journal of Social Robotics (SORO): 2018, 2019

## **Grant Reviewing**

NSF Panelist: 2016(x2), 2018, 2019, 2020, 2021, 2022

## **University Service**

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### **College Level**

- Faculty Hiring Committee for Whole Communities Whole Health Cluster Hires: 2019–2020
- CNS Fall Lab Working Group in response to COVID restrictions: 2020

### **Department Level**

- Texas Robotics Machine Shop Committee: 2020–2022
- Texas Robotics Space Committee: 2020–2022
- UTCS Turing Scholars Admissions Committee: 2020–2022
- UTCS Diversity, Equity, and Inclusion Committee: 2020–2022
- UTCS Graduate Admissions Committee: 2019–2020
- UMass CICS Honors Program Director: 2018–2019
- UMass CICS Undergraduate Course Assistant Program Director: 2018–2019
- UMass CICS Graduate Admissions Committee: 2015–2016
- UMass CICS Student Activities Committee: 2015–2016
- UMass CICS Data Science Faculty Hiring Committee: 2016–2017
- UMass CICS Student Activities Committee: 2016–2017

## **Advising and Thesis Committees**

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### **Postdoctoral Supervisor**

- Rohan Chandra, 2022 – 2024, currently Assistant Professor at the University of Virginia
- Kiarash (Kia) Rahmani, 2022 – 2024, currently Applied AI Scientist at Durable AI
- Nathan Tsoi, 2025 – present

## **PhD Supervisor**

- Sarah Etter, UT Austin. 2023–present
- Dongmyeong Lee, UT Austin. 2023–present
- Cheng-Chun Hsu, UT Austin. 2023–present
- Arthur Zhang, UT Austin. 2022–present
- Zichao Hu, UT Austin. 2022–present
- Sadanand Modak, UT Austin. 2022–present
- Noah Patton (Co-advised by Isil Dillig), UT Austin. 2022–present
- Eric Hsiung (Co-advised by Swarat Chaudhuri), UT Austin. 2022–present
- Amanda Adkins, UT Austin. 2020–present
- Emily Pruc, UMass Amherst. 2018–2022
- Sadegh Rabiee, UT Austin. 2016–2022, currently at Amazon Lab126  
*Introspective Perception for Mobile Robots*
- Jarrett Holtz, UT Austin. 2015–2022, currently at Bosch Research  
*Leveraging Program Synthesis for Robust Long-Term Robot Autonomy via Interactive Learning and Adaptation*
- Samer Nashed (Changed advisors in 2019), UMass Amherst. 2015–2019
- Spencer Lane (Changed advisors in 2019), UMass Amherst. 2016–2019
- Alyxander Burns (Changed advisors in 2019), UMass Amherst. 2017–2019

## **Master’s Thesis Supervisor**

- Kavan Sikand, UT Austin. 2019–2022
- David Balaban, UMass Amherst, 2016 – 2018  
*A Real-Time Solver For Time-Optimal Control Of Omnidirectional Robots with Bounded Acceleration*

## **Undergraduate Honors Thesis Supervisor**

- Elvin Yang, UT Austin. 2021–2022  
*Wait, That Feels Familiar: Learning to Extrapolate Human Preferences for Preference-Aligned Path Planning*  
**Best Honors Thesis Award**
- Rahul Menon, UT Austin. 2021–2022  
*Terrain-Adaptive Global Planning from Local Demonstrations*
- Shakeel Samsudeen, UT Austin. 2021–2022  
*Context-Aware Object SLAM*

- Nathaniel Plaxton, UT Austin. 2021–2022  
*Estimating Kinodynamic Uncertainty Using Learned Gaussian Noise Models*
- Michael Satanovski, UT Austin, 2021-2022  
*An Empirical Evaluation of LIDAR Object Detectors for Autonomous Mobile Robots*
- Edward Schneeweiss, UMass Amherst, 2015 – 2019  
*Joint Perception and Planning for Obstacle Avoidance over Non-Planar Terrain*
- Kyle Vedder, UMass Amherst, 2015 – 2019  
*X\*: Anytime Multiagent Path Planning With Bounded Search*
- George Larionov, UMass Amherst, 2015 – 2016  
*Human-robot Interaction: Integrating Speech Recognition with a Mobile Robot System*

### **PhD Committee Member**

- Connor Basich, UMass Amherst. Supervisor: Shlomo Zilberstein
- Minkyu Kim, UT Austin. Supervisor: Luis Sentis
- Abhinav Verma, UT Austin. Supervisor: Swarat Chaudhuri
- Kyle Hollins Wray, UMass Amherst. Supervisor: Shlomo Zilberstein
- Justin Svegliato, UMass Amherst. Supervisor: Shlomo Zilberstein
- Tiffany Liu, UMass Amherst. Supervisor: Roderic Grupen
- Takeshi Takahashi, UMass Amherst. Supervisor: Roderic Grupen
- Mike Lanighan, UMass Amherst. Supervisor: Roderic Grupen
- Keen Sung, UMass Amherst. Supervisor: Brian Levine
- (Thesis Opponent) <sup>1</sup>, Nils Bore, KTH. Supervisor: John Folkesson

### **Undergraduate Honors Thesis Committee Member**

- Stefan Kussmaul, UMass Amherst. Supervisor: Roderic Grupen
- Karl Schmeckpepper, UMass Amherst. Supervisor: Roderic Grupen

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<sup>1</sup> A PhD thesis dissertation in the Swedish doctoral system is formally presented by an external examiner, called the *thesis opponent*. A thesis opponent places the work of the PhD thesis in context with the state of the art, presents the findings of the thesis, and leads a discussion with questions.

## Publications

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### Conference Papers

- [1] Taijing Chen, Sateesh Kumar, Junhong Xu, Georgios Pavlakos, Joydeep Biswas, and Roberto Martín-Martín. “Searching in Space and Time: Unified Memory-Action Loops for Open-World Object Retrieval.” In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: [https://joydeepb.com/Publications/icra2026\\_memory\\_action.pdf](https://joydeepb.com/Publications/icra2026_memory_action.pdf).
- [2] Hochul Hwang, Soowan Yang, Jahir Sadik Monon, Nicholas A. Giudice, Sunghoon Ivan Lee, Joydeep Biswas, and Donghyun Kim. “GuideNav: User-Informed Development of a Vision-Only Robotic Navigation Assistant for Blind Travelers.” In: *Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction*. 2026, pp. 1129–1139. DOI: 10.1145/3757279.3785625. URL: [https://joydeepb.com/Publications/hri2026\\_guidenav.pdf](https://joydeepb.com/Publications/hri2026_guidenav.pdf).
- [3] Hochul Hwang, Soowan Yang, Anh H. N. Nguyen, Parth Goel, Krisha Adhikari, Sunghoon Ivan Lee, Joydeep Biswas, Nicholas A. Giudice, and Donghyun Kim. “GuideTWSI: A Diverse Tactile Walking Surface Indicator Dataset from Synthetic and Real-World Images for Blind and Low-Vision Navigation.” In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: [https://joydeepb.com/Publications/icra2026\\_guidetwsi.pdf](https://joydeepb.com/Publications/icra2026_guidetwsi.pdf).
- [4] Dongmyeong Lee, Jesse Quattrociochi, Christian Ellis, Rwik Rana, Amanda Adkins, Adam Uccello, Garrett Warnell, and Joydeep Biswas. “BEV-Patch-PF: Particle Filtering with BEV-Aerial Feature Matching for Off-Road Geo-Localization.” In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: <https://arxiv.org/pdf/2512.15111>.
- [5] Wei Liu, Huihua Zhao, Chenran Li, Yuchen Deng, Joydeep Biswas, Soha Pouya, and Yan Chang. “COMPASS: Cross-Embodiment Mobility Policy via Residual RL and Skill Synthesis.” In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: [https://joydeepb.com/Publications/arxiv2025\\_compass.pdf](https://joydeepb.com/Publications/arxiv2025_compass.pdf).
- [6] Rwik Rana, Jesse Quattrociochi, Dongmyeong Lee, Christian Ellis, Amanda Adkins, Adam Uccello, Garrett Warnell, and Joydeep Biswas. “OverSeeC: Open-Vocabulary CostMap Generation from Satellite Images and Natural Language.” In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: <https://arxiv.org/pdf/2602.18606>.
- [7] Nathan Tsoi, Eric Hsiung, Masayuki Yamazaki, Swarat Chaudhuri, Peter Stone, and Joydeep Biswas. “Coadaptive Value Alignment.” In: *Proceedings of the 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS) Blue Sky Ideas Track*. 2026. DOI: 10.65109/naik4475. URL: [https://joydeepb.com/Publications/aamas2026\\_coadaptive.pdf](https://joydeepb.com/Publications/aamas2026_coadaptive.pdf).
- [8] Zifan Xu, Kristen Procko, Michael Munje, Kristin Patterson, Lea Sabatini, Joydeep Biswas, and Peter Stone. “The Essentials of AI for Life and Society: A Full-Scale AI Literacy Course Accessible to All.” In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 40. 47. 2026, pp. 40730–40736. DOI: 10.1609/aaai.v40i47.41521. URL: [https://joydeepb.com/Publications/eaai2026\\_essentials.pdf](https://joydeepb.com/Publications/eaai2026_essentials.pdf).
- [9] Zifan Xu, Myoungkyu Seo, Dongmyeong Lee, Hao Fu, Jiaheng Hu, Jiaxun Cui, Yuqian Jiang, Zhihan Wang, Anastasiia Brund, Joydeep Biswas, and Peter Stone. “Learning Agile Striker Skills for Humanoid Soccer Robots from Noisy Sensory Input.” In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: [https://joydeepb.com/Publications/icra2026\\_striker.pdf](https://joydeepb.com/Publications/icra2026_striker.pdf).

- [10] Shangqun Yu, Hochul Hwang, Trung M. Dang, Joydeep Biswas, Nicholas A. Giudice, Sunghoon Ivan Lee, and Donghyun Kim. “Human-Centered Development of Guide Dog Robots: Quiet and Stable Locomotion Control.” In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: [https://joydeepb.com/Publications/arxiv2025\\_guide\\_dog\\_hcd.pdf](https://joydeepb.com/Publications/arxiv2025_guide_dog_hcd.pdf).
- [11] Arthur Zhang, Xiangyun Meng, Luca Calliari, Dong-Ki Kim, Shayegan Omidshafiei, Joydeep Biswas, Ali Agha, and Amirreza Shaban. “VENTURA: Adapting Image Diffusion Models for Unified Task Conditioned Navigation.” In: *2026 IEEE International Conference on Robotics and Automation (ICRA)*. 2026. URL: [https://joydeepb.com/Publications/arxiv2025\\_ventura.pdf](https://joydeepb.com/Publications/arxiv2025_ventura.pdf).
- [12] Abrar Anwar, John Welsh, Joydeep Biswas, Soha Pouya, and Yan Chang. “ReMEmbR: Building and Reasoning Over Long-Horizon Spatio-Temporal Memory for Robot Navigation.” In: *2025 IEEE International Conference on Robotics and Automation (ICRA)*. 2025. DOI: 10.1109/ICRA55743.2025.11127706. URL: <https://arxiv.org/pdf/2409.13682>.
- [13] Joydeep Biswas, Don Fussell, Peter Stone, Kristin Patterson, Kristen Procko, Lea Sabatini, and Zifan Xu. “The Essentials of AI for Life and Society: An AI Literacy Course for the University Community.” In: *Educational Advances in Artificial Intelligence*. Vol. 39. 28. 2025, pp. 28973–28978. DOI: 10.1609/aaai.v39i28.35166. URL: [https://joydeepb.com/Publications/eaai2025\\_essentials.pdf](https://joydeepb.com/Publications/eaai2025_essentials.pdf).
- [14] Rohan Chandra, Haresh Karnan, Negar Mehr, Peter Stone, and Joydeep Biswas. “Multi-Agent Inverse Reinforcement Learning in Real World Unstructured Pedestrian Crowds.” In: *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2025, pp. 18668–18675. DOI: 10.1109/IROS60139.2025.11247542. URL: [https://joydeepb.com/Publications/iros2025\\_multi.pdf](https://joydeepb.com/Publications/iros2025_multi.pdf).
- [15] Rohit Dwivedula, Sadanand Modak, Aditya Akella, Joydeep Biswas, Daehyeok Kim, and Christopher J. Rossbach. “ConfigBot: Adaptive Resource Allocation for Robot Applications in Dynamic Environments.” In: *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. 2025, pp. 10792–10799. DOI: 10.1109/IROS60139.2025.11245932. URL: [https://joydeepb.com/Publications/iros2025\\_configbot.pdf](https://joydeepb.com/Publications/iros2025_configbot.pdf).
- [16] Eric Hsiung, Joydeep Biswas, and Swarat Chaudhuri. “Automata Learning from Preference and Equivalence Queries.” In: *Proceedings of the 37th International Conference on Computer Aided Verification (CAV 2025)*. Vol. 15934. Springer, 2025, pp. 104–126. DOI: 10.1007/978-3-031-98685-7\_5. URL: [https://joydeepb.com/Publications/cav2025\\_remap.pdf](https://joydeepb.com/Publications/cav2025_remap.pdf).
- [17] Cheng-Chun Hsu, Bowen Wen, Jie Xu, Yashraj Narang, Xiaolong Wang, Yuke Zhu, Joydeep Biswas, and Stan Birchfield. “SPOT: SE(3) Pose Trajectory Diffusion for Object-Centric Manipulation.” In: *2025 IEEE International Conference on Robotics and Automation (ICRA)*. 2025. DOI: 10.1109/ICRA55743.2025.11127562. URL: [https://joydeepb.com/Publications/icra2025\\_spot.pdf](https://joydeepb.com/Publications/icra2025_spot.pdf).
- [18] Zichao Hu, Junyi Jessy Li, Arjun Guha, and Joydeep Biswas. “Robo-Instruct: Simulator-Augmented Instruction Alignment For Finetuning Code LLMs.” In: *Conference on Language Modeling (CoLM)*. 2025. URL: [https://joydeepb.com/Publications/colm2025\\_roboinstruct.pdf](https://joydeepb.com/Publications/colm2025_roboinstruct.pdf).
- [19] Zichao Hu, Chen Tang, Michael Joseph Munje, Yifeng Zhu, Alex Liu, Shuijing Liu, Garrett Warnell, Peter Stone, and Joydeep Biswas. “ComposableNav: Instruction-Following Navigation in Dynamic Environments via Composable Diffusion.” In: *Conference on Robot Learning (CoRL)*. 2025. URL: [https://joydeepb.com/Publications/corl2025\\_composablenav.pdf](https://joydeepb.com/Publications/corl2025_composablenav.pdf).

- [20] Jiahui Huang, Qunjie Zhou, Hesam Rabeti, Aleksandr Korovko, Huan Ling, Xuanchi Ren, Tianchang Shen, Jun Gao, Dmitry Slepichev, Chen-Hsuan Lin, Jiawei Ren, Kevin Xie, Joydeep Biswas, Laura Leal-Taixe, and Sanja Fidler. “ViPE: Video Pose Engine for 3D Geometric Perception.” In: *NVIDIA Research Whitepapers*. 2025. URL: [https://joydeepb.com/Publications/nvidia2025\\_vipe.pdf](https://joydeepb.com/Publications/nvidia2025_vipe.pdf).
- [21] Sangmin Jung, Utkarsh Nath, Yezhou Yang, Giulia Pedrielli, Joydeep Biswas, Amy Zhang, Hassan Ghasemzadeh, and Pavan Turaga. “Guiding Diffusion with Deep Geometric Moments: Balancing Fidelity and Variation.” In: *CVPR 3rd Workshop on Generative Models for Computer Vision (GMCV)*. 2025. URL: [https://joydeepb.com/Publications/cvprw2025\\_geometric\\_moments.pdf](https://joydeepb.com/Publications/cvprw2025_geometric_moments.pdf).
- [22] Wei Liu, Huihua Zhao, Chenran Li, Joydeep Biswas, Billy Okal, Pulkit Goyal, Soha Pouya, and Yan Chang. “X-Mobility: End-To-End Generalizable Navigation via World Modeling.” In: *2025 IEEE International Conference on Robotics and Automation (ICRA)*. 2025. DOI: 10.1109/ICRA55743.2025.11128692. URL: [https://joydeepb.com/Publications/icra2025\\_xmobility.pdf](https://joydeepb.com/Publications/icra2025_xmobility.pdf).
- [23] Sadanand Modak, Noah Tobias Patton, Isil Dillig, and Joydeep Biswas. “SYNAPSE: SYmbolic Neural-Aided Preference Synthesis Engine.” In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 39. 26. 2025, pp. 27529–27537. DOI: 10.1609/aaai.v39i26.34965. URL: [https://joydeepb.com/Publications/aaai2025\\_synapse.pdf](https://joydeepb.com/Publications/aaai2025_synapse.pdf).
- [24] Michael Joseph Munje, Chen Tang, Shuijing Liu, Zichao Hu, Yifeng Zhu, Jiaxun Cui, Garrett Warnell, Joydeep Biswas, and Peter Stone. “SocialNav-SUB: Benchmarking VLMs for Scene Understanding in Social Robot Navigation.” In: *Conference on Robot Learning (CoRL)*. 2025. URL: [https://joydeepb.com/Publications/corl2025\\_socialnavsub.pdf](https://joydeepb.com/Publications/corl2025_socialnavsub.pdf).
- [25] Zixuan Wu, Francesca Lucchetti, Aleksander Boruch-Gruszecki, Jingmiao Zhao, Carolyn Jane Anderson, Joydeep Biswas, Federico Cassano, and Arjun Guha. “ReasoningWeekly: A General Knowledge and Verbal Reasoning Challenge for Large Language Models.” In: *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*. 2025, pp. 3124–3140. DOI: 10.18653/v1/2025.ijcnlp-long.167. URL: [https://joydeepb.com/Publications/ijcnlp2025\\_reasoningweekly.pdf](https://joydeepb.com/Publications/ijcnlp2025_reasoningweekly.pdf).
- [26] Arthur Zhang, Harshit Sikchi, Amy Zhang, and Joydeep Biswas. “CREStE: Scalable Mapless Navigation with Internet Scale Priors and Counterfactual Guidance.” In: *Proceedings of Robotics: Science and Systems*. 2025. DOI: 10.15607/RSS.2025.XXI.136. URL: [https://joydeepb.com/Publications/rss2025\\_creste.pdf](https://joydeepb.com/Publications/rss2025_creste.pdf).
- [27] Joydeep Biswas. “Teaching Robots to ”Get It Right”.” In: *Proceedings of the 37th International FLAIRS Conference*. 2024. DOI: 10.32473/flairs.37.1.135854. URL: <https://joydeepb.com/Publications/flairs2024teaching.pdf>.
- [28] Rohan Chandra, Zayne Sprague, and Joydeep Biswas. “SOCIALGYM 2.0: Simulator for Multi-Robot Learning and Navigation in Shared Human Spaces.” In: *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 38. 21. 2024, pp. 23778–23780. DOI: 10.1609/aaai.v38i21.30562. URL: <https://arxiv.org/pdf/2303.05584.pdf>.
- [29] Elliott Hauser, Yao-Cheng Chan, Sadanand Modak, Joydeep Biswas, and Justin Hart. “Vid2Real HRI: Align Video-Based HRI Study Designs with Real-World Settings.” In: *2024 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*. 2024, pp. 542–548. DOI: 10.1109/RO-MAN60168.2024.10731413. URL: [https://joydeepb.com/Publications/arxiv2024\\_vid2real\\_hri.pdf](https://joydeepb.com/Publications/arxiv2024_vid2real_hri.pdf).

- [30] Hochul Hwang, Hee-Tae Jung, Nicholas A Giudice, Joydeep Biswas, Sunghoon Ivan Lee, and Donghyun Kim. “Towards Robotic Companions: Understanding Handler-Guide Dog Interactions for Informed Guide Dog Robot Design.” In: *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*. New York, NY, USA: Association for Computing Machinery, 2024. DOI: 10 . 1145/3613904.3642181. URL: [https://joydeepb.com/Publications/chi2024\\_guidedogrobots.pdf](https://joydeepb.com/Publications/chi2024_guidedogrobots.pdf).
- [31] Haresh Karnan, Elvin Yang, Garrett Warnell, Joydeep Biswas, and Peter Stone. “Wait, That Feels Familiar: Learning to Extrapolate Human Preferences for Preference Aligned Path Planning.” In: *2024 IEEE International Conference on Robotics and Automation (ICRA)*. 2024. DOI: 10 . 1109/ICRA57147.2024.10611475. URL: [https://joydeepb.com/Publications/icra2024\\_wait.pdf](https://joydeepb.com/Publications/icra2024_wait.pdf).
- [32] Yung-Chi Kung, Arthur Zhang, Junmin Wang, and Joydeep Biswas. “Looking Inside Out: Anticipating Driver Intent from Videos.” In: *2024 IEEE International Conference on Robotics and Automation (ICRA)*. 2024. DOI: 10 . 1109/ICRA57147.2024.10610257. URL: [https://joydeepb.com/Publications/icra2024\\_looking.pdf](https://joydeepb.com/Publications/icra2024_looking.pdf).
- [33] Luisa Mao, Ryan Soussan, Brian Coltin, Trey Smith, and Joydeep Biswas. “Semantic Masking and Visual Feature Matching for Robust Localization.” In: *2024 International Conference on Space Robotics (iSpaRo)*. 2024, pp. 1–7. DOI: 10 . 1109/ISPARO60631.2024.10688001. URL: [https://joydeepb.com/Publications/isparo2024\\_semantic\\_masking.pdf](https://joydeepb.com/Publications/isparo2024_semantic_masking.pdf).
- [34] Amir Hossain Raj, Zichao Hu, Haresh Karnan, Rohan Chandra, Amirreza Payandeh, Luisa Mao, Peter Stone, Joydeep Biswas, and Xuesu Xiao. “Rethinking Social Robot Navigation: Leveraging the Best of Two Worlds.” In: *2024 IEEE International Conference on Robotics and Automation (ICRA)*. 2024. DOI: 10 . 1109/ICRA57147.2024.10611710. URL: [https://joydeepb.com/Publications/icra2024\\_rethinking.pdf](https://joydeepb.com/Publications/icra2024_rethinking.pdf).
- [35] Elliott Hauser, Yao-Cheng Chan, Parth Chonkar, Geethika Hemkumar, Huihai Wang, Daksh Dua, Shikhar Gupta, Efren Mendoza Enriquez, Tiffany Kao, Justin Hart, Reuth Mirsky, Joydeep Biswas, Junfeng Jiao, and Peter Stone. ““What’s That Robot Doing Here?”: Perceptions Of Incidental Encounters With Autonomous Quadruped Robots.” In: *Proceedings of the First International Symposium on Trustworthy Autonomous Systems*. TAS ’23. Edinburgh, United Kingdom: Association for Computing Machinery, 2023. DOI: 10 . 1145/3597512.3599707. URL: <https://joydeepb.com/Publications/tas2023incidental.pdf>.
- [36] Hochul Hwang, Tim Xia, Ibrahima Keita, Ken Suzuki, Joydeep Biswas, Sunghoon I. Lee, and Donghyun Kim. “System Configuration and Navigation of a Guide Dog Robot: Toward Animal Guide Dog-Level Guiding Work.” In: *2023 IEEE International Conference on Robotics and Automation (ICRA)*. 2023, pp. 9778–9784. DOI: 10 . 1109/ICRA48891.2023.10160573. URL: [https://joydeepb.com/Publications/icra2023\\_guide\\_dog.pdf](https://joydeepb.com/Publications/icra2023_guide_dog.pdf).
- [37] Haresh Karnan, Elvin Yang, Daniel Farkash, Garrett Warnell, Joydeep Biswas, and Peter Stone. “STERLING: Self-Supervised Terrain Representation Learning from Unconstrained Robot Experience.” In: *Proceedings of The 7th Conference on Robot Learning*. Ed. by Jie Tan, Marc Toussaint, and Kourosh Darvish. Vol. 229. Proceedings of Machine Learning Research. PMLR, 2023, pp. 2393–2413. URL: [https://joydeepb.com/Publications/corl2023\\_sterling.pdf](https://joydeepb.com/Publications/corl2023_sterling.pdf).
- [38] Joshua Vekhter and Joydeep Biswas. “Responsible Robotics: A Socio-Ethical Addition To Robotics Courses.” In: *Educational Advances in Artificial Intelligence*. 2023, pp. 15877–15885. DOI: 10 . 1609/aaai.v37i13.26885. URL: [https://joydeepb.com/Publications/eaai2023\\_responsible\\_robotics.pdf](https://joydeepb.com/Publications/eaai2023_responsible_robotics.pdf).
- [39] Amanda Adkins, Taijing Chen, and Joydeep Biswas. “Probabilistic Object Maps for Long-Term Robot Localization.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE, 2022, pp. 931–938. DOI: 10 . 1109/IROS47612.2022.9981316. URL: [https://joydeepb.com/Publications/iros2022\\_pom.pdf](https://joydeepb.com/Publications/iros2022_pom.pdf).



- [40] Pranav Atreya and Joydeep Biswas. “State Supervised Steering Function for Sampling-based Kinodynamic Planning.” In: *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. International Foundation for Autonomous Agents and Multiagent Systems. 2022, pp. 35–43. URL: [https://joydeepb.com/Publications/aamas2022\\_s3f.pdf](https://joydeepb.com/Publications/aamas2022_s3f.pdf).
- [41] Pranav Atreya, Haresh Karnan, Kavan Singh Sikand, Xuesu Xiao, Sadegh Rabiee, and Joydeep Biswas. “High-Speed Accurate Robot Control using Learned Forward Kinodynamics and Non-linear Least Squares Optimization.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE. 2022, pp. 11789–11795. DOI: 10.1109/IROS47612.2022.9981259. URL: [https://joydeepb.com/Publications/iros2022\\_optimfk.pdf](https://joydeepb.com/Publications/iros2022_optimfk.pdf).
- [42] Jarrett Holtz and Joydeep Biswas. “SOCIALGYM: A Framework for Benchmarking Social Robot Navigation.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE. 2022, pp. 11246–11252. DOI: 10.1109/IROS47612.2022.9982021. URL: [https://joydeepb.com/Publications/iros2022\\_socialgym.pdf](https://joydeepb.com/Publications/iros2022_socialgym.pdf).
- [43] Haresh Karnan, Kavan Singh Sikand, Pranav Atreya, Sadegh Rabiee, Xuesu Xiao, Garrett Warnell, Peter Stone, and Joydeep Biswas. “VI-IKD: High-Speed Accurate Off-Road Navigation using Learned Visual-Inertial Inverse Kinodynamics.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE. 2022, pp. 3294–3301. DOI: 10.1109/IROS47612.2022.9982060. URL: [https://joydeepb.com/Publications/iros2022\\_viikd.pdf](https://joydeepb.com/Publications/iros2022_viikd.pdf).
- [44] Kavan Singh Sikand, Sadegh Rabiee, Adam Uccello, Xuesu Xiao, Garrett Warnell, and Joydeep Biswas. “Visual Representation Learning for Preference-Aware Path Planning.” In: *Robotics and Automation (ICRA), IEEE International Conference on*. 2022, pp. 11303–11309. DOI: 10.1109/ICRA46639.2022.9811828. URL: [https://joydeepb.com/Publications/icra2022\\_vrlpap.pdf](https://joydeepb.com/Publications/icra2022_vrlpap.pdf).
- [45] Jiayi Wei, Jarrett Holtz, Isil Dillig, and Joydeep Biswas. “STEADY: Simultaneous State Estimation and Dynamics Learning from Indirect Observations.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE. 2022, pp. 6593–6599. DOI: 10.1109/IROS47612.2022.9981279. URL: [https://joydeepb.com/Publications/iros2022\\_steady.pdf](https://joydeepb.com/Publications/iros2022_steady.pdf).
- [46] Jarrett Holtz, Simon Andrews, Arjun Guha, and Joydeep Biswas. “Iterative Program Synthesis for Adaptable Social Navigation.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. 2021, pp. 6256–6261. DOI: 10.1109/IROS51168.2021.9636540. URL: [https://joydeepb.com/Publications/iros2021\\_idips.pdf](https://joydeepb.com/Publications/iros2021_idips.pdf).
- [47] Kavan Singh Sikand, Logan Zartman, Sadegh Rabiee, and Joydeep Biswas. “Robofleet: Secure Open Source Communication and Management for Fleets of Autonomous Robots.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. 2021, pp. 406–412. DOI: 10.1109/IROS51168.2021.9635830. URL: [https://joydeepb.com/Publications/iros2021\\_robofleet.pdf](https://joydeepb.com/Publications/iros2021_robofleet.pdf).
- [48] Jiayi Wei, Tongrui Li, Swarat Chaudhuri, Isil Dillig, and Joydeep Biswas. “OneVision: Centralized to Distributed Controller Synthesis with Delay Compensation.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. 2021, pp. 398–405. DOI: 10.1109/IROS51168.2021.9636164. URL: [https://joydeepb.com/Publications/iros2021\\_onevision.pdf](https://joydeepb.com/Publications/iros2021_onevision.pdf).
- [49] Connor Basich, Justin Svegliato, Kyle Hollins Wray, Stefan Witwicki, Joydeep Biswas, and Shlomo Zilberstein. “Learning to Optimize Autonomy in Competence-Aware Systems.” In: *Proceedings of the 2020 international conference on Autonomous agents and multi-agent systems*. International Foundation for Autonomous Agents and Multiagent Systems. 2020, pp. 123–131. DOI: 10.5555/3398761.3398781. URL: [https://joydeepb.com/Publications/aamas2020\\_cas.pdf](https://joydeepb.com/Publications/aamas2020_cas.pdf).

- [50] Jarrett Holtz, Arjun Guha, and Joydeep Biswas. “Robot Action Selection Learning via Layered Dimension Informed Program Synthesis.” In: *Conference on Robot Learning*. 2020, pp. 1471–1480. URL: [https://joydeepb.com/Publications/corl2020\\_ldips.pdf](https://joydeepb.com/Publications/corl2020_ldips.pdf).
- [51] Sadeqh Rabiee and Joydeep Biswas. “TV-SLAM: Introspective Vision for Simultaneous Localization and Mapping.” In: *Conference on Robot Learning*. 2020, pp. 1100–1109. URL: [https://joydeepb.com/Publications/corl2020\\_ivslam.pdf](https://joydeepb.com/Publications/corl2020_ivslam.pdf).
- [52] Joseph Spitzer, Joydeep Biswas, and Arjun Guha. “Making High-Performance Robots Safe and Easy to Use for an Introduction to Computing.” In: *Educational Advances in Artificial Intelligence*. 2020, pp. 13469–13476. DOI: 10.1609/aaai.v34i09.7065. URL: [https://joydeepb.com/Publications/eaai2020\\_jsbots.pdf](https://joydeepb.com/Publications/eaai2020_jsbots.pdf).
- [53] Joydeep Biswas. “The Quest For ”Always-On” Autonomous Mobile Robots.” In: *Proceedings of the Twenty-Eighth International Joint Conference on Artificial Intelligence, IJCAI-19*. International Joint Conferences on Artificial Intelligence Organization, July 2019, pp. 6388–6392. DOI: 10.24963/ijcai.2019/893. URL: [https://joydeepb.com/Publications/ijcai2019\\_early\\_career\\_spotlight.pdf](https://joydeepb.com/Publications/ijcai2019_early_career_spotlight.pdf).
- [54] Sadeqh Rabiee and Joydeep Biswas. “A Friction-Based Kinematic Model for Skid-Steer Wheeled Mobile Robots.” In: *Robotics and Automation (ICRA), IEEE International Conference on*. IEEE. 2019, pp. 8563–8569. DOI: 10.1109/ICRA.2019.8794216. URL: [https://joydeepb.com/Publications/icra2019\\_skid\\_steer.pdf](https://joydeepb.com/Publications/icra2019_skid_steer.pdf).
- [55] Sadeqh Rabiee and Joydeep Biswas. “IVOA: Introspective Vision for Obstacle Avoidance.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE. 2019, pp. 1230–1235. DOI: 10.1109/IROS40897.2019.8968176. URL: [https://joydeepb.com/Publications/iros2019\\_ivoa.pdf](https://joydeepb.com/Publications/iros2019_ivoa.pdf).
- [56] Justin Svegliato, Kyle Hollins Wray, Stefan J. Witwicki, Joydeep Biswas, and Shlomo Zilberstein. “Belief Space Metareasoning for Exception Recovery.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE. 2019, pp. 1224–1229. DOI: 10.1109/IROS40897.2019.8967676. URL: [https://joydeepb.com/Publications/iros2019\\_belief.pdf](https://joydeepb.com/Publications/iros2019_belief.pdf).
- [57] Kyle Vedder and Joydeep Biswas. “X\*: Anytime Multiagent Path Planning With Bounded Search.” In: *Proc. of the 18th International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*. 2019, pp. 2247–2249. ISBN: 9781450363099. URL: [https://joydeepb.com/Publications/aamas2019\\_xastar.pdf](https://joydeepb.com/Publications/aamas2019_xastar.pdf).
- [58] David Balaban, Alexander Fischer, and Joydeep Biswas. “A Real-Time Solver For Time-Optimal Control Of Omnidirectional Robots with Bounded Acceleration.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. 2018, pp. 8027–8032. DOI: 10.1109/IROS.2018.8594306. URL: [https://joydeepb.com/Publications/iros2018\\_tsocs.pdf](https://joydeepb.com/Publications/iros2018_tsocs.pdf).
- [59] Jarrett Holtz, Arjun Guha, and Joydeep Biswas. “Interactive Robot Transition Repair With SMT.” In: *International Joint Conference on Artificial Intelligence (IJCAI)*. 2018, pp. 4905–4911. DOI: 10.24963/ijcai.2018/681. URL: [https://joydeepb.com/Publications/ijcai2018\\_srtr.pdf](https://joydeepb.com/Publications/ijcai2018_srtr.pdf).
- [60] Samer Nashed and Joydeep Biswas. “Human-in-the-Loop SLAM.” In: *AAAI Conference on Artificial Intelligence*. 2018, pp. 1503–1510. URL: [https://joydeepb.com/Publications/aaai2018\\_hitl-slam.pdf](https://joydeepb.com/Publications/aaai2018_hitl-slam.pdf).
- [61] Samer Nashed, David Ilstrup, and Joydeep Biswas. “Localization under Topological Uncertainty for Lane Identification of Autonomous Vehicles.” In: *Robotics and Automation (ICRA), IEEE International Conference on*. 2018, pp. 6000–6005. URL: [https://joydeepb.com/Publications/icra2018\\_lutu.pdf](https://joydeepb.com/Publications/icra2018_lutu.pdf).

- [62] Sourish Ghosh and Joydeep Biswas. “Joint Perception And Planning For Efficient Obstacle Avoidance Using Stereo Vision.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE. 2017, pp. 1026–1031. URL: [https://joydeepb.com/Publications/iros2017\\_jpp.pdf](https://joydeepb.com/Publications/iros2017_jpp.pdf).
- [63] Jarrett Holtz and Joydeep Biswas. “Automatic Extrinsic Calibration of Depth Sensors with Ambiguous Environments and Restricted Motion.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE. 2017, pp. 2235–2240. URL: [https://joydeepb.com/Publications/delta\\_calibration.pdf](https://joydeepb.com/Publications/delta_calibration.pdf).
- [64] Juan Pablo Mendoza, Joydeep Biswas, Philip Cooksey, Richard Wang, Steven Klee, Danny Zhu, and Manuela Veloso. “Selectively Reactive Coordination for a Team of Robot Soccer Champions.” In: *AAAI Conference on Artificial Intelligence*. 2016, pp. 3354–3360. URL: <https://joydeepb.com/Publications/aaai2016selectively.pdf>.
- [65] Samer Nashed and Joydeep Biswas. “Curating Long-Term Vector Maps.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE. 2016, pp. 4643–4648. DOI: 10.1109/IROS.2016.7759683. URL: [https://joydeepb.com/Publications/iros2016\\_ltv.pdf](https://joydeepb.com/Publications/iros2016_ltv.pdf).
- [66] Priyam Parashar, Robert Fisher, Reid Simmons, Manuela Veloso, and Joydeep Biswas. “Learning Context-Based Outcomes for Mobile Robots in Unstructured Indoor Environments.” In: *2015 IEEE 14th International Conference on Machine Learning and Applications (ICMLA)*. 2015, pp. 703–706. DOI: 10.1109/ICMLA.2015.222. URL: <https://joydeepb.com/Publications/icmla2015learning.pdf>.
- [67] Manuela Veloso, Joydeep Biswas, Brian Coltin, and Stephanie Rosenthal. “CoBots: Robust symbiotic autonomous mobile service robots.” In: *Proceedings of the 24th International Conference on Artificial Intelligence*. AAAI Press. 2015, pp. 4423–4429. URL: [https://joydeepb.com/Publications/ijcai2015\\_cobots.pdf](https://joydeepb.com/Publications/ijcai2015_cobots.pdf).
- [68] Alfredo Weitzenfeld, Joydeep Biswas, Mehmet Akar, and Kanjanapan Sukvichai. “RoboCup Small-Size League: Past, Present and Future.” In: *RoboCup 2014: Robot World Cup XVIII*. Springer International Publishing, 2015, pp. 611–623. URL: [https://joydeepb.com/Publications/robocup2014\\_ssl.pdf](https://joydeepb.com/Publications/robocup2014_ssl.pdf).
- [69] Danny Zhu, Joydeep Biswas, and Manuela Veloso. “AutoRef: Towards Real-Robot Soccer Complete Automated Refereeing.” In: *RoboCup 2014: Robot World Cup XVIII*. Springer International Publishing, 2015, pp. 419–430. URL: [https://joydeepb.com/Publications/robocup2014\\_autoref.pdf](https://joydeepb.com/Publications/robocup2014_autoref.pdf).
- [70] Joydeep Biswas, Juan Pablo Mendoza, Danny Zhu, Benjamin Choi, Steven Klee, and Manuela Veloso. “Opponent-driven planning and execution for pass, attack, and defense in a multi-robot soccer team.” In: *Proceedings of the 2014 international conference on Autonomous agents and multi-agent systems*. International Foundation for Autonomous Agents and Multiagent Systems. 2014, pp. 493–500. URL: [https://joydeepb.com/Publications/aamas2014\\_cmdragons.pdf](https://joydeepb.com/Publications/aamas2014_cmdragons.pdf).
- [71] Joydeep Biswas and Manuela Veloso. “Episodic Non-Markov localization: reasoning about short-term and long-term features.” In: *Robotics and Automation (ICRA), 2014 IEEE International Conference on*. IEEE. 2014, pp. 3969–3974. DOI: 10.1109/ICRA.2014.6907435. URL: [https://joydeepb.com/Publications/icra2014\\_enml.pdf](https://joydeepb.com/Publications/icra2014_enml.pdf).
- [72] Joydeep Biswas and Manuela Veloso. “Model-Instance Object Mapping.” In: *RoboCup 2014: Robot World Cup XVIII*. Springer International Publishing, 2014, pp. 525–536. URL: [https://joydeepb.com/Publications/robocup2014\\_object\\_maps.pdf](https://joydeepb.com/Publications/robocup2014_object_maps.pdf).
- [73] Joydeep Biswas and Manuela Veloso. “Multi-sensor mobile robot localization for diverse environments.” In: *RoboCup 2013: Robot World Cup XVII*. Springer Berlin Heidelberg, 2014, pp. 468–479. URL: [https://joydeepb.com/Publications/13robocup\\_multisensor.pdf](https://joydeepb.com/Publications/13robocup_multisensor.pdf).

- [74] Stefan Zickler, Tim Laue, José Angelo Gurzoni Jr, Oliver Birbach, Joydeep Biswas, and Manuela Veloso. “Five Years of SSL-Vision–Impact and Development.” In: *RoboCup 2013: Robot World Cup XVII*. Springer Berlin Heidelberg, 2014, pp. 656–663. URL: [https://joydeepb.com/Publications/robocup2013\\_ssl-vision.pdf](https://joydeepb.com/Publications/robocup2013_ssl-vision.pdf).
- [75] Benjamin Choi, Cetin Meriçli, Joydeep Biswas, and Manuela Veloso. “Fast human detection for indoor mobile robots using depth images.” In: *Robotics and Automation (ICRA), 2013 IEEE International Conference on*. IEEE. 2013, pp. 1108–1113. DOI: 10.1109/ICRA.2013.6630711. URL: [https://joydeepb.com/Publications/icra2013\\_human\\_detection.pdf](https://joydeepb.com/Publications/icra2013_human_detection.pdf).
- [76] Joydeep Biswas and Manuela Veloso. “Depth camera based indoor mobile robot localization and navigation.” In: *Robotics and Automation (ICRA), 2012 IEEE International Conference on*. IEEE. 2012, pp. 1697–1702. DOI: 10.1109/ICRA.2012.6224766. URL: [https://joydeepb.com/Publications/icra2012\\_kinectLocalization.pdf](https://joydeepb.com/Publications/icra2012_kinectLocalization.pdf).
- [77] Joydeep Biswas and Manuela Veloso. “Planar polygon extraction and merging from depth images.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE. 2012, pp. 3859–3864. DOI: 10.1109/IROS.2012.6385841. URL: [https://joydeepb.com/Publications/iros2012\\_planes.pdf](https://joydeepb.com/Publications/iros2012_planes.pdf).
- [78] Brian Coltin, Joydeep Biswas, Dean Pomerleau, and Manuela Veloso. “Effective semi-autonomous telepresence.” In: *RoboCup 2011: Robot Soccer World Cup XV*. Springer Berlin Heidelberg, 2012, pp. 365–376. URL: <https://joydeepb.com/Publications/11robocup-telepresence.pdf>.
- [79] Manuela Veloso, Joydeep Biswas, Brian Coltin, Stephanie Rosenthal, Tom Kollar, Cetin Mericli, Mehdi Samadi, Susana Brandao, and Rodrigo Ventura. “CoBots: Collaborative robots servicing multi-floor buildings.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE. 2012, pp. 5446–5447. DOI: 10.1109/IROS.2012.6386300. URL: [https://joydeepb.com/Publications/iros2012\\_cobots.pdf](https://joydeepb.com/Publications/iros2012_cobots.pdf).
- [80] Joydeep Biswas, Brian Coltin, and Manuela Veloso. “Corrective gradient refinement for mobile robot localization.” In: *Intelligent Robots and Systems (IROS), IEEE/RSJ International Conference on*. IEEE. 2011, pp. 73–78. DOI: 10.1109/IROS.2011.6094625. URL: [https://joydeepb.com/Publications/iros2011\\_cgr.pdf](https://joydeepb.com/Publications/iros2011_cgr.pdf).
- [81] Joydeep Biswas and Manuela Veloso. “Wifi localization and navigation for autonomous indoor mobile robots.” In: *Robotics and Automation (ICRA), 2010 IEEE International Conference on*. IEEE. 2010, pp. 4379–4384. DOI: 10.1109/ROBOT.2010.5509842. URL: [https://joydeepb.com/Publications/icra2010\\_wifi.pdf](https://joydeepb.com/Publications/icra2010_wifi.pdf).
- [82] Stephanie Rosenthal, Joydeep Biswas, and Manuela Veloso. “An effective personal mobile robot agent through symbiotic human-robot interaction.” In: *Proceedings of the 9th International Conference on Autonomous Agents and Multiagent Systems: volume 1-Volume 1*. International Foundation for Autonomous Agents and Multiagent Systems. 2010, pp. 915–922. URL: [https://joydeepb.com/Publications/Rosenthal\\_AAMAS2010.pdf](https://joydeepb.com/Publications/Rosenthal_AAMAS2010.pdf).
- [83] Brian C Becker, Cristina Robles Valdivieso, Joydeep Biswas, Louis Lobes, Cameron N Riviere, et al. “Active guidance for laser retinal surgery with a handheld instrument.” In: *Engineering in Medicine and Biology Society, 2009. EMBC 2009. Annual International Conference of the IEEE*. IEEE. 2009, pp. 5587–5590. DOI: 10.1109/IEMBS.2009.5333489. URL: [https://joydeepb.com/Publications/Becker\\_et\\_al\\_Micron.pdf](https://joydeepb.com/Publications/Becker_et_al_Micron.pdf).

## Journal Articles

- [84] Shuijing Liu, Haochen Xia, Fatemeh Cheraghi Pouria, Kaiwen Hong, Neeloy Chakraborty, Zichao Hu, Joydeep Biswas, and Katherine Driggs-Campbell. “HEIGHT: Heterogeneous Interaction Graph Transformer for Robot Navigation in Crowded and Constrained Environments.” In: *IEEE Transactions on Automation Science and Engineering* 23 (2026), pp. 1211–1230. DOI: 10.1109/TASE.2025.3646588. URL: [https://joydeepb.com/Publications/tase2025\\_height.pdf](https://joydeepb.com/Publications/tase2025_height.pdf).
- [85] Amanda Adkins, Dongmyeong Lee, and Joydeep Biswas. “CLOVER: Context-Aware Long-Term Object Viewpoint- and Environment-Invariant Representation Learning.” In: *IEEE Robotics and Automation Letters* 10.11 (2025), pp. 11928–11935. DOI: 10.1109/LRA.2025.3613991. URL: [https://joydeepb.com/Publications/icra2026\\_clover.pdf](https://joydeepb.com/Publications/icra2026_clover.pdf).
- [86] Rohan Chandra, Vrushabh Zinage, Efstathios Bakolas, Peter Stone, and Joydeep Biswas. “Deadlock-free, Safe, and Decentralized Multi-Robot Navigation in Social Mini-Games via Discrete-Time Control Barrier Functions.” In: *Autonomous Robots* 49.2 (2025), p. 12. DOI: 10.1007/s10514-025-10194-8. URL: [https://joydeepb.com/Publications/autonomousrobots2025\\_deadlock.pdf](https://joydeepb.com/Publications/autonomousrobots2025_deadlock.pdf).
- [87] Anthony Francis, Claudia Pérez-d’Arpino, Chengshu Li, Fei Xia, Alexandre Alahi, Rachid Alami, Aniket Bera, Abhijat Biswas, Joydeep Biswas, Rohan Chandra, Hao-Tien Lewis Chiang, Michael Everett, Sehoon Ha, Justin Hart, Jonathan P. How, Haresh Karnan, Tsang-Wei Edward Lee, Luis J. Manso, Reuth Mirksy, Soeren Pirk, Phani Teja Singamaneni, Peter Stone, Ada V. Taylor, Peter Trautman, Nathan Tsoi, Marynel Vazquez, Xuesu Xiao, Peng Xu, Naoki Yokoyama, Alexander Toshev, and Roberto Martin-Martin. “Principles and Guidelines for Evaluating Social Robot Navigation Algorithms.” In: *ACM Transactions on Human-Robot Interaction* 14.2 (2025), pp. 1–65. DOI: 10.1145/3700599. URL: [https://joydeepb.com/Publications/thri2025\\_principles.pdf](https://joydeepb.com/Publications/thri2025_principles.pdf).
- [88] Luisa Mao, Garrett Warnell, Peter Stone, and Joydeep Biswas. “PACER: Preference-Conditioned All-Terrain Costmap Generation.” In: *IEEE Robotics and Automation Letters* 10.5 (2025), pp. 4572–4579. DOI: 10.1109/LRA.2025.3549645. URL: [https://joydeepb.com/Publications/ral2025\\_pacer.pdf](https://joydeepb.com/Publications/ral2025_pacer.pdf).
- [89] Amanda Adkins, Taijing Chen, and Joydeep Biswas. “ObVi-SLAM: Long-Term Object-Visual SLAM.” In: *IEEE Robotics and Automation Letters* 9.3 (2024), pp. 2909–2916. DOI: 10.1109/LRA.2024.3363534. URL: [https://joydeepb.com/Publications/ral2024\\_obvislam.pdf](https://joydeepb.com/Publications/ral2024_obvislam.pdf).
- [90] Arko Banerjee, Kia Rahmani, Joydeep Biswas, and Isil Dillig. “Dynamic model predictive shielding for provably safe reinforcement learning.” In: *Advances in Neural Information Processing Systems* 37 (2024), pp. 100131–100159. DOI: 10.5555/3737916.3741093. URL: [https://joydeepb.com/Publications/neurips2024\\_dmps.pdf](https://joydeepb.com/Publications/neurips2024_dmps.pdf).
- [91] Johannes Betz, Hongrui Zheng, Felix Jahncke, Zirui Zang, Florian Sauerbeck, Y. Rosa Zheng, Joydeep Biswas, Venkat Krovi, and Rahul Mangharam. “FITENTH: Enhancing Autonomous Systems Education Through Hands-On Learning and Competition.” In: *IEEE Transactions on Intelligent Vehicles* (2024). DOI: 10.1109/TIV.2024.3495227. URL: <https://ieeexplore.ieee.org/document/10749837>.
- [92] Zichao Hu, Francesca Lucchetti, Claire Schlesinger, Yash Saxena, Anders Freeman, Sadanand Modak, Arjun Guha, and Joydeep Biswas. “Deploying and Evaluating LLMs to Program Service Mobile Robots.” In: *IEEE Robotics and Automation Letters* 9.3 (2024), pp. 2853–2860. DOI: 10.1109/LRA.2024.3360020. URL: [https://joydeepb.com/Publications/ral2024\\_codebotler.pdf](https://joydeepb.com/Publications/ral2024_codebotler.pdf).

- [93] Noah Patton, Kia Rahmani, Meghana Missula, Joydeep Biswas, and Isil Dillig. “Programming-by-Demonstration for Long-Horizon Robot Tasks.” In: *Proceedings of the ACM on Programming Languages* 8.POPL (2024), pp. 512–545. URL: [https://joydeepb.com/Publications/popl2024\\_prolex.pdf](https://joydeepb.com/Publications/popl2024_prolex.pdf).
- [94] Jimmy Xin, Linus Zheng, Kia Rahmani, Jiayi Wei, Jarrett Holtz, Isil Dillig, and Joydeep Biswas. “Programmatic Imitation Learning From Unlabeled and Noisy Demonstrations.” In: *IEEE Robotics and Automation Letters* 9.6 (2024), pp. 4894–4901. DOI: 10.1109/LRA.2024.3385691. URL: [https://joydeepb.com/Publications/ral2024\\_plunder.pdf](https://joydeepb.com/Publications/ral2024_plunder.pdf).
- [95] Arthur Zhang, Chaitanya Eranki, Christina Zhang, Ji-Hwan Park, Raymond Hong, Pranav Kalyani, Lochana Kalyanaraman, Arsh Gamare, Arnav Bagad, Maria Esteva, and Joydeep Biswas. “Toward Robust Robot 3-D Perception in Urban Environments: The UT Campus Object Dataset.” In: *IEEE Transactions on Robotics* 40 (2024), pp. 3322–3340. DOI: 10.1109/TRO.2024.3400831. URL: [https://joydeepb.com/Publications/tro2024\\_coda.pdf](https://joydeepb.com/Publications/tro2024_coda.pdf).
- [96] Connor Basich, Justin Svegliato, Kyle H Wray, Stefan Witwicki, Joydeep Biswas, and Shlomo Zilberstein. “Competence-aware systems.” In: *Artificial Intelligence* 316 (2023), p. 103844. URL: <https://joydeepb.com/Publications/aij2023cas.pdf>.
- [97] Rohan Chandra, Rahul Maligi, Arya Anantula, and Joydeep Biswas. “SocialMAPF: Optimal and Efficient Multi-Agent Path Finding With Strategic Agents for Social Navigation.” In: *IEEE Robotics and Automation Letters* (2023), pp. 3214–3221. DOI: 10.1109/LRA.2023.3265169. URL: <https://joydeepb.com/Publications/ral2023socialmapf.pdf>.
- [98] Sadeqh Rabiee and Joydeep Biswas. “Introspective Perception for Mobile Robots.” In: *Artificial Intelligence* 324 (2023), p. 103999. ISSN: 0004-3702. DOI: <https://doi.org/10.1016/j.artint.2023.103999>. URL: <https://joydeepb.com/Publications/aij2023introspective.pdf>.
- [99] Divyanshu Saxena, Nihal Sharma, Donghyun Kim, Rohit Dwivedula, Jiayi Chen, Chenxi Yang, Sri-ram Ravula, Zichao Hu, Aditya Akella, Sebastian Angel, et al. “On a Foundation Model for Operating Systems.” In: *arXiv preprint arXiv:2312.07813* (2023). URL: <https://arxiv.org/pdf/2312.07813.pdf>.
- [100] Haresh Karnan, Anirudh Nair, Xuesu Xiao, Garrett Warnell, Soeren Pirk, Alexander Toshev, Justin Hart, Joydeep Biswas, and Peter Stone. “Socially CompliAnt Navigation Dataset (SCAND): A Large-Scale Dataset Of Demonstrations For Social Navigation.” In: *IEEE Robotics and Automation Letters* 7.4 (2022), pp. 11807–11814. DOI: 10.1109/LRA.2022.3184025. URL: [https://joydeepb.com/Publications/ral2022\\_scand.pdf](https://joydeepb.com/Publications/ral2022_scand.pdf).
- [101] Sadeqh Rabiee, Connor Basich, Kyle Hollins Wray, Shlomo Zilberstein, and Joydeep Biswas. “Competence-Aware Path Planning via Introspective Perception.” In: *IEEE Robotics and Automation Letters* (2022), pp. 3218–3225. DOI: 10.1109/LRA.2022.3145517. URL: [https://joydeepb.com/Publications/ral2022\\_cpip.pdf](https://joydeepb.com/Publications/ral2022_cpip.pdf).
- [102] Xuesu Xiao, Zifan Xu, Zizhao Wang, Yunlong Song, Garrett Warnell, Peter Stone, Tingnan Zhang, Shravan Ravi, Gary Wang, Haresh Karnan, et al. “Autonomous Ground Navigation in Highly Constrained Spaces: Lessons Learned From the Benchmark Autonomous Robot Navigation Challenge at ICRA 2022 [Competitions].” In: *IEEE Robotics & Automation Magazine* 29.4 (2022), pp. 148–156. DOI: 10.1109/MRA.2022.3213466. URL: <https://joydeepb.com/Publications/ram2022barn.pdf>.
- [103] Kyle Vedder and Joydeep Biswas. “X\*: Anytime Multi-Agent Path Finding For Sparse Domains Using Iterative Repairs.” In: *Artificial Intelligence* 291 (2021), p. 103417. DOI: 10.1016/j.artint.2020.103417. URL: [https://joydeepb.com/Publications/aij\\_xstar.pdf](https://joydeepb.com/Publications/aij_xstar.pdf).

- [104] Xuesu Xiao, Joydeep Biswas, and Peter Stone. “Learning Inverse Kinodynamics for Accurate High-Speed Off-Road Navigation on Unstructured Terrain.” In: *IEEE Robotics and Automation Letters (RA-L)* (2021), pp. 1–7. DOI: 10.1109/LRA.2021.3090023. URL: <https://joydeepb.com/Publications/ral2021ikd.pdf>.
- [105] Jonathan Aldrich, David Garlan, Christian Kästner, Claire Le Goues, Anahita Mohseni-Kabir, Ivan Ruchkin, Selva Samuel, Bradley Schmerl, Christopher Steven Timperley, Manuela Veloso, et al. “Model-based adaptation for robotics software.” In: *IEEE Software* 36.2 (2019), pp. 83–90. URL: [https://joydeepb.com/Publications/ieee\\_mars.pdf](https://joydeepb.com/Publications/ieee_mars.pdf).
- [106] Keen Yuun Sung, Joydeep Biswas, Erik G. Learned-Miller, Brian Neil Levine, and Marc Liberator. “Server-side Traffic Analysis Reveals Mobile Location Information over the Internet.” In: *IEEE Transactions on Mobile Computing* (2018). DOI: 10.1109/TMC.2018.2857777. URL: [https://joydeepb.com/Publications/tmc2018\\_sung.pdf](https://joydeepb.com/Publications/tmc2018_sung.pdf).
- [107] Joydeep Biswas and Manuela M. Veloso. “Episodic non-Markov localization.” In: *Robotics and Autonomous Systems* 87 (2017), pp. 162–176. ISSN: 0921-8890. DOI: 10.1016/j.robot.2016.09.005. URL: [https://joydeepb.com/Publications/ras\\_episodic\\_nonmarkov\\_localization.pdf](https://joydeepb.com/Publications/ras_episodic_nonmarkov_localization.pdf).
- [108] Joydeep Biswas and Manuela Veloso. “The 1,000-km Challenge: Insights and Quantitative and Qualitative Results.” In: *IEEE Intelligent Systems* 31.3 (2016), pp. 86–96. DOI: 10.1109/MIS.2016.53. URL: [https://joydeepb.com/Publications/intelligent\\_systems2016\\_1000km.pdf](https://joydeepb.com/Publications/intelligent_systems2016_1000km.pdf).
- [109] Joydeep Biswas and Manuela M Veloso. “Localization and navigation of the CoBots over long-term deployments.” In: *The International Journal of Robotics Research* 32.14 (2013), pp. 1679–1694. DOI: 10.1177/0278364913503892. URL: [https://joydeepb.com/Publications/ijrr\\_longterm\\_autonomy\\_cobot.pdf](https://joydeepb.com/Publications/ijrr_longterm_autonomy_cobot.pdf).
- [110] Joydeep Biswas and B Seth. “Dynamic stabilisation of a reaction-wheel actuated wheel-robot.” In: *International Journal of Factory Automation, Robotics and Soft Computing* 4 (2008), pp. 96–101. URL: [https://www.joydeepb.com/Publications/Biswas\\_Seth\\_Wheel\\_Robot.pdf](https://www.joydeepb.com/Publications/Biswas_Seth_Wheel_Robot.pdf).

## **Other Publications**

- [111] Joydeep Biswas, Sheila Schoepp, Gautham Vasan, Anthony Opipari, Arthur Zhang, Zichao Hu, Sebastian Joseph, Matthew Lease, Junyi Jessy Li, Peter Stone, Kiri L. Wagstaff, Matthew E. Taylor, and Odest Chadwicke Jenkins. *AI-Assisted Peer Review at Scale: The AAAI-26 AI Review Pilot*. arXiv preprint arXiv:2604.13940. 2026. DOI: 10.48550/arXiv.2604.13940. URL: [https://joydeepb.com/Publications/arxiv2026\\_ai\\_peer\\_review.pdf](https://joydeepb.com/Publications/arxiv2026_ai_peer_review.pdf).
- [112] Zichao Hu, Zifan Xu, Dongsik Chang, He Yin, Linh Tran, Roberto Martín-Martín, Peter Stone, Jingyu Qiao, and Joydeep Biswas. *VOFA: Visual Object Goal Pushing with Force-Adaptive Control for Humanoids*. arXiv preprint arXiv:2605.01518. 2026. DOI: 10.48550/arXiv.2605.01518. URL: [https://joydeepb.com/Publications/arxiv2026\\_vofa.pdf](https://joydeepb.com/Publications/arxiv2026_vofa.pdf).
- [113] Aleksander Boruch-Gruszecki, Yangtian Zi, Zixuan Wu, Tejas Oberoi, Carolyn Jane Anderson, Joydeep Biswas, and Arjun Guha. *Agnostics: Learning to Code in Any Programming Language via Reinforcement with a Universal Learning Environment*. 2025. arXiv: 2508.04865 [cs.LG]. URL: <https://arxiv.org/abs/2508.04865>.
- [114] Alexander Korovko, Dmitry Slepichev, Alexander Efitarov, Aigul Dzhumamuratova, Viktor Kuznetsov, Hesam Rabeti, Joydeep Biswas, and Soha Pouya. *cuVSLAM: CUDA Accelerated Visual Odometry and Mapping*. arXiv preprint arXiv:2506.04359. 2025. DOI: 10.48550/arXiv.2506.04359. URL: [https://joydeepb.com/Publications/arxiv2025\\_cuvslam.pdf](https://joydeepb.com/Publications/arxiv2025_cuvslam.pdf).

- [115] Luisa Mao, Garrett Warnell, Peter Stone, and Joydeep Biswas. *Terrain Costmap Generation via Scaled Preference Conditioning*. arXiv preprint arXiv:2511.11529. 2025. DOI: 10.48550/arXiv.2511.11529. URL: [https://joydeepb.com/Publications/arxiv2025\\_spacer.pdf](https://joydeepb.com/Publications/arxiv2025_spacer.pdf).
- [116] Jingrui Yu, Jun Liu, Kefei Ren, Joydeep Biswas, Rurui Ye, Keqiang Wu, Chirag Majithia, and Di Zeng. *CuSfM: CUDA-Accelerated Structure-from-Motion*. arXiv preprint arXiv:2510.15271. 2025. DOI: 10.48550/arXiv.2510.15271. URL: [https://joydeepb.com/Publications/arxiv2025\\_cusfm.pdf](https://joydeepb.com/Publications/arxiv2025_cusfm.pdf).
- [117] Eric Hsiung, Swarat Chaudhuri, and Joydeep Biswas. *Learning Quantitative Automata Modulo Theories*. arXiv preprint arXiv:2411.10601. 2024. DOI: 10.48550/arXiv.2411.10601. URL: [https://joydeepb.com/Publications/arxiv2024\\_quant\\_automata.pdf](https://joydeepb.com/Publications/arxiv2024_quant_automata.pdf).
- [118] Claire Schlesinger, Arjun Guha, and Joydeep Biswas. *Creating and Repairing Robot Programs in Open-World Domains*. arXiv preprint arXiv:2410.18893. 2024. DOI: 10.48550/arXiv.2410.18893. URL: [https://joydeepb.com/Publications/arxiv2024\\_roborepair.pdf](https://joydeepb.com/Publications/arxiv2024_roborepair.pdf).
- [119] Yunhao Yang, Neel P. Bhatt, William Ward, Zichao Hu, Joydeep Biswas, and Ufuk Topcu. *Joint Verification and Refinement of Language Models for Safety-Constrained Planning*. arXiv preprint arXiv:2410.14865. 2024. DOI: 10.48550/arXiv.2410.14865. URL: [https://joydeepb.com/Publications/arxiv2024\\_joint\\_verification.pdf](https://joydeepb.com/Publications/arxiv2024_joint_verification.pdf).
- [120] Eric Hsiung, Joydeep Biswas, and Swarat Chaudhuri. *Learning Reward Machines through Preference Queries over Sequences*. 2023. arXiv: 2308.09301 [cs.LG]. URL: <https://arxiv.org/pdf/2308.09301.pdf>.
- [121] Bo Liu, Yuqian Jiang, Xiaohan Zhang, Qiang Liu, Shiqi Zhang, Joydeep Biswas, and Peter Stone. *LLM+P: Empowering Large Language Models with Optimal Planning Proficiency*. 2023. arXiv: 2304.11477 [cs.AI]. URL: <https://arxiv.org/pdf/2304.11477.pdf>.
- [122] Asha Kailin Jain, Maxwell Svetlik, Nicholas Machak, Kavan Singh Sikand, Cem Karamanli, Kaiyu Zhou, Justin Hart, Joydeep Biswas, Luis Sentis, and Junfeng Jiao. *An Open-Source Framework for Last Mile Delivery with Heterogeneous Robots*. AAAI Spring Symposium Series, Machine Learning for Mobile Robot Navigation in the Wild. 2021. URL: [https://joydeepb.com/Publications/aaaisss2021\\_smads.pdf](https://joydeepb.com/Publications/aaaisss2021_smads.pdf).
- [123] Jenna Claire Hammond, Joydeep Biswas, and Arjun Guha. *Automatic Failure Recovery for End-User Programs on Service Mobile Robots*. arXiv Preprint arXiv:1909.02778. 2019. URL: [https://joydeepb.com/Publications/arxiv\\_rtpl.pdf](https://joydeepb.com/Publications/arxiv_rtpl.pdf).
- [124] Jarrett Holtz, Arjun Guha, and Joydeep Biswas. *SMT-based Robot Transition Repair*. arXiv Preprint arXiv:2001.04397. 2019. URL: [https://joydeepb.com/Publications/arxiv\\_srtr.pdf](https://joydeepb.com/Publications/arxiv_srtr.pdf).
- [125] Spencer Lane, Kyle Vedder, and Joydeep Biswas. *Augmenting Planning Graphs in 2-Dimensional Dynamic Environments With Obstacle Scaffolds*. PlanRob Workshop, International Conference on Automated Planning and Scheduling. 2017. URL: [https://joydeepb.com/Publications/planrob2017\\_scaffold.pdf](https://joydeepb.com/Publications/planrob2017_scaffold.pdf).
- [126] Joydeep Biswas. *An Enclosed One Wheel Robot*. B.Tech Project Final Report, Institute of Technology, Bombay, India. 2008. URL: <https://joydeepb.com/Publications/btp2008joydeepb.pdf>.



## **PhD Thesis**

- [127] Joydeep Biswas. “Vector Map-Based, Non-Markov Localization for Long-Term Deployment of Autonomous Mobile Robots.” PhD Thesis. Carnegie Mellon University, Dec. 2014. URL: [https://joydeepb.com/Publications/joydeepb\\_thesis.pdf](https://joydeepb.com/Publications/joydeepb_thesis.pdf).